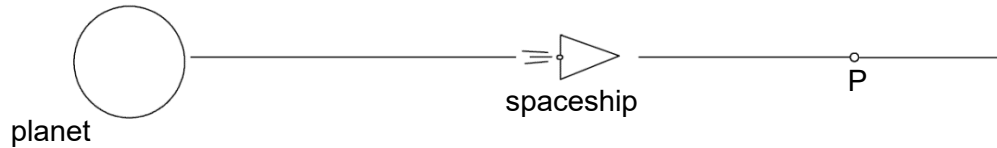
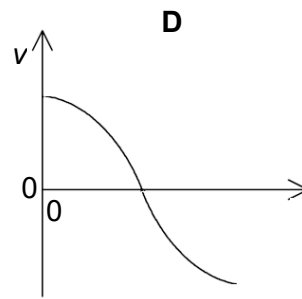
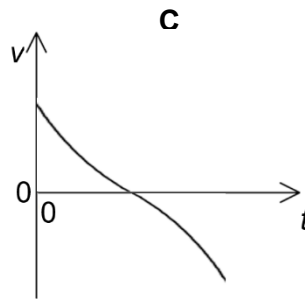
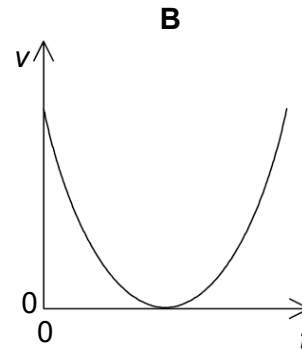
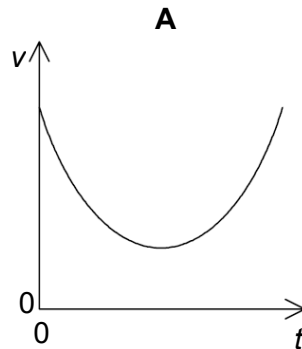


- 10 A powered spaceship is moving directly away from a planet as shown below.



The spaceship passes point P at $t = 0$. At point P, the thrusters of the spaceship are switched off but the spaceship remains under the influence of the planet. Which one of the following graphs best represents the subsequent variation with time t of the velocity v of the spaceship?



Ans: C

Gradient is equal to acceleration, as well as gravitational field strength. As the spaceship moves away from the planet, gravitational field strength must decrease. After the horizontal intercept where $v = 0$, the spaceship will be travelling toward the planet.

- 11 A satellite of mass 200 kg is moved from an orbit where the gravitational potential is